

Summary

GPU-focused computational scientist and software engineer for performance-oriented systems and HPC roles. Builds Python and C++ code with CUDA, ROCm/HIP, MPI, and OpenMP; sharded JAX/PyTorch computational graphs across up to 800 Frontier GPUs and TB-scale datasets.

Technical Skills

Programming: Python, C, C++, CUDA, HIP, JAX, PyTorch, Bash

GPU / HPC: CUDA, ROCm/HIP, cuBLAS, cuTENSOR, cuRAND, rocBLAS, rocFFT, MPI, OpenMP, collective communication, multi-GPU parallelization, PETSc, SLEPc

Platforms: OLCF Frontier (AMD MI250X), NERSC Perlmutter (NVIDIA A100), ALCF Aurora (Intel GPUs), Linux, CMake, Docker, Kubernetes

Debug / Perf: cuda-gdb, nvprof, TotalView, Linaro DDT, MPI/GPU distributed debugging, strong scaling, weak scaling, hardware-aware optimization

Machine Learning: PyTorch, PyTorch Geometric, automatic differentiation, equivariant graph neural networks, scikit-learn

Experience

Yale University, Electrical Engineering

PhD Research Assistant

2021–present

- Implemented INCITE-supported distributed scientific software in Python/C++ for LiF and double-perovskite exciton-polaron workloads, using MPI, OpenMP, CUDA/ROCm/HIP, PETSc, SLEPc, and GPU linear-algebra libraries on Frontier and Perlmutter.
- Parallelized matrix and vector workloads across multiple GPUs, using collective communication and distributed linear-algebra workflows to scale simulations across clusters.
- Ported and debugged scientific workloads across OLCF Frontier (AMD MI250X), NERSC Perlmutter (NVIDIA A100), and ALCF Aurora (Intel GPU) environments, translating many-body and excited-state methods into scalable implementations.
- Built DeePH-style computational graphs and PyTorch Geometric equivariant models that delivered roughly 100x speedups for parts of the workflow by combining training with automatic differentiation to obtain additional quantities without separate expensive calculations.
- Sharded JAX/PyTorch computational graphs across up to 800 Frontier GPUs and TB-scale datasets; ran strong- and weak-scaling studies using cuda-gdb, nvprof, TotalView, CUDA-aware MPI workflows, and Linaro DDT to debug multi-process and multi-GPU applications.
- Packaged scientific software in Docker containers for reproducible deployment across local and HPC systems.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS research in 2024, 2025, and 2026, grounding GPU and HPC work in production research workflows.

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed CUDA-based software for high-volume acoustic waveform processing, visualization, and field-tool diagnostics in geological well logging systems.
- Built engineering software and supporting backend tooling for real-time remote tool testing, data collection, and sensor calibration workflows.
- Applied finite-element analysis with dolfinx for frequency analysis of steel pipes in well environments to support physics-driven engineering studies.

FindLight

Photonics Technical Writer

2018–2019

- Wrote clear technical content on lasers, optics, and photonics devices for engineering audiences and product-focused readers.

Selected Projects

Multi-GPU Exciton-Phonon Simulation: Implemented Quantum ESPRESSO and post-DFT many-body workflows for LiF and double-perovskite exciton-polarons; ran CUDA/ROCm, MPI, and OpenMP matrix/vector parallelization on up to 800 GPUs and TB-scale data.

Scientific ML on GPU Systems: Developed PyTorch Geometric equivariant models that produced roughly 100x workflow speedups by pairing training with automatic differentiation to recover additional quantities efficiently.

CUDA Containers for HPC Software: Packaged scientific applications in CUDA-ready Docker environments for portable

deployment across local and HPC systems.

Matrix-Multiply AI Accelerator: Built a Verilog-based matrix-multiply accelerator project for AI-oriented datapath design and hardware-aware compute fundamentals.

Education

Yale University <i>PhD, Electrical Engineering</i>	<i>2021–present</i>
Yale University <i>MPhil, Electrical Engineering</i>	<i>2023</i>
Yale University <i>MS, Electrical Engineering</i>	<i>2023</i>
Cornell University <i>MEng, Electrical and Computer Engineering</i>	<i>2017–2018</i>
Cornell University <i>BS, Electrical and Computer Engineering</i>	<i>2013–2017</i>